

ANTON ALYAKIN

alyakin314@gmail.com $\diamond$ alyakin314.github.io

EDUCATION

Washington University in St. Louis

Doctor of Medicine

2024 Congress of Neurological Surgeons AI Fellow

[Aug 2021, May 2027]

(Expected)

Johns Hopkins University

Master of Science in Engineering

Applied Mathematics & Statistics

[Jan 2019, Dec 2019]

Johns Hopkins University

Bachelor of Science

Computer Science

Applied Mathematics & Statistics

[Aug 2015, May 2019]

RESEARCH

NYU Langone Health

AI Research Engineer

Neurosurgery / OLAB

[Mar 2024, May 2026]

Washington University in Saint Louis

Medical Student Researcher

Neurosurgery / Leuthardt Lab

[Mar 2023, May 2026]

Johns Hopkins University

Assistant Research Engineer

Applied Mathematics & Statistics / Neurodata Lab

[Jan 2020, Mar 2021]

Graduate Research Assistant

Applied Mathematics & Statistics / Priebe Group

[Jun 2019, Dec 2019]

Undergraduate Research Assistant

Computer Science / Saria Lab

[May 2017, May 2019]

SKILLS

Languages (in order of proficiency):

Python, English, Russian, R, Matlab.

Python skills:

PyTorch[DDP, FSDP, DeepSpeed, PyTorchLightning, LightningFabric, HuggingFaceAccelerate],
Tensorflow, vLLM, W&B, API[OpenAI, Anthropic, AWS Bedrock].

Other skills:

L^AT_EX, Git, Docker, Databases[PostgreSQL, BigQuery], Agentic Engineering[ClaudeCode, Codex],
React[Next.js, Tailwind], HPC[IBM Spectrum LSF, SLURM], Platforms[Heroku, Fly.io, Vercel],
Lead climbing (6b/5.10), Bouldering (7a/V6 indoor; V2 outdoor).

DISSERTATIONS

1. **A. Alyakin**, *Robust Hypothesis Testing of Location Parameters using Lq-Likelihood-Ratio-Type Test in Python*, a thesis submitted to The Johns Hopkins University in conformity with the requirements for the degree of Master of Science in Engineering, 2019. [arXiv] [library] [code]
2. **A. Alyakin**, *Motif Discovery in the Irregularly Sampled Time Series Data*, a thesis submitted to The Johns Hopkins University in conformity with the requirements for Senior Honors Thesis in Computer Science, 2019. [osf] [pdf]

PUBLICATIONS

1. **A. Alyakin**, E. K. Oermann, *Current Artificial Intelligence: Large Language Model Applications in Neurosurgery*, Neurosurgery, accepted.
2. A. Kapoor, **A. Alyakin**, J. E. Markert, A. Arias, M. Sughrue, E. Yang, K. Vishwanath, J. V. Lee, E. K. Oermann, *Big Data in U.S. Neuro-Oncology: Trends and Translational Priorities*, Journal of Neuro-Oncology, 2026. [journal]
3. K. Vishwanath, **A. Alyakin**, M. Ghosh, A. Hage, S. N. Neifert, C. Orillac, N. Mandelberg, H. Khan, J. V. Lee, J. Yao, W. R. Small, A. Varma, B. Hewitt, Y. Aphinyanaphongs, D. A. Alber, E. K. Oermann, *General-Purpose Large Language Models Outperform Specialized Clinical AI Tools on Medical Benchmarks*, Nature Medicine, 2026. [journal] [arXiv] [code]
4. **A. Alyakin**, J. Stryker, D. A. Alber, J. V. Lee, K. L. Sangwon, B. Duderstadt, A. Save, D. Kurland, S. Frome, S. Singh, J. Zhang, E. Yang, K. Y. Park, C. Orillac, A. Valliani, S. Neifert, A. Liu, A. Patel, C. Livia, D. Lau, I. Laufer, P. A. Rozman, E. T. Hidalgo, H. Riina, R. Feng, T. Hollon, Y. Aphinyanaphongs, J. G. Golfinos, L. Snyder, E. Leuthardt, D. Kondziolka, E. K. Oermann, *CNS-Obsidian: A Neurosurgical Vision-Language Model Built From Scientific Publications*, Neurosurgery, 2026. [journal] [arXiv] [code] [model]
5. **A. Alyakin**, D. A. Alber, J. V. Lee, J. Stryker, S. Singh, A. Save, D. Kurland, C. Orillac, A. A. Valliani, S. Neifert, D. Lau, I. Laufer, P. A. Rozman, E. T. Hidalgo, H. Riina, L. Snyder, E. C. Leuthardt, D. Kondziolka, E. K. Oermann, *Automated Generation and Human Evaluation of Neurosurgical Board Examination Self-Assessment Questions*, Neurosurgery Practice, 2026.
6. **A. Alyakin**, J. Stryker, J. V. Lee, R. Feng, T. Hollon, D. Kondziolka, E. K. Oermann, *AI-Powered Pipeline Transforms Neurosurgical Articles into High-Quality Graphical Abstracts*, Neurosurgery Practice, 2026. [journal]
7. K. L. Sangwon, X. Han, A. Becker, Y. Zhang, R. Ni, J. Zhang, D. A. Alber, **A. Alyakin**, M. Nakatsuka, N. Fabbri, Y. Aphinyanaphongs, J. Yang, A. Chachoua, D. Kondziolka, I. Laufer, E. K. Oermann, *Automating the Referral of Bone Metastases Patients With and Without the Use of Large Language Models*, Neurosurgery, 2026. [journal]
8. N. I. Maarouf, D. Reinecke, A. Smith, J. Markert, T. G. Cogan, X. Han, **A. Alyakin**, D. A. Alber, M. Park, N. K. Goff, H. Weiss, E. Harake, K. Eddy, T. Hollon, E. K. Oermann, D. A. Orringer, *Natural Language Processing Methods Automate Molecular Marker Extraction From Glioma Pathology Reports*, Neurosurgery, 2026. [journal]
9. S. F. Chen, **A. Alyakin**, A. Seas, E. Yang, J. Choi, J. V. Lee, A. Chen, P. Warman, R. Bitolas, R. J. Steele, D. Alber, E. K. Oermann, *LLM-assisted Systematic Review of Large Language Models in Clinical Medicine*, Nature Medicine, 2026. [journal] [code]
10. K. Vishwanath, **A. Alyakin**, M. Ghosh, J. V. Lee, D. A. Alber, K. Sangwon, D. Kondziolka, E. K. Oermann, *Evaluating the Performance and Fragility of Large Language Models on Neurosurgical Board-like Questions*, Neurosurgery, 2025. [journal] [arXiv]
11. X. Han^{†1}, **A. Alyakin**[†], S. Ciprut, C. Lapierre, J. Stryker, J. Golfinos, D. Kondziolka, E. K. Oermann, *Neuro Data Hub: A New Approach for Streamlining Medical Clinical Research*, Neurosurgery Practice, 2025. [journal]
12. K. Vishwanath, J. Stryker, **A. Alyakin**, D. A. Alber, E. K. Oermann, *MedMobile: a Mobile-sized Language Model with Clinical Capabilities*, BMJ Digital Health & AI, 2025. [journal] [arXiv] [code]
13. S. Singh, **A. Alyakin**, D. A. Alber, J. Stryker, A. P. S. Tong, K. Sangwon, N. Goff, M. de la Paz, M. Hernandez-Rovira, K. Y. Park, D. Kondziolka, E. K. Oermann, *The Pitfalls of Multiple-Choice Questions in Generative AI and Medical Education*, Scientific Reports, 2025. [journal] [arXiv]

^{1†} symbolizes equal contribution; order preserved as in manuscript.

14. **A. Alyakin**, D. Kurland, D. Alber, K. Sangwon, D. Li, A. Tsirigos, E. Leuthardt, D. Kondziolka, E. Oermann, *CNS-CLIP: Transforming a Neurosurgical Journal into a Multimodal Medical Model*, Neurosurgery, 2025. [journal]
15. D. A. Alber, Z. Yang, **A. Alyakin**, E. Yang, J. Zhang, S. Rai, A. A. Valliani, G. R. Rosenbaum, A. K. Amend-Thomas, D. B. Kurland, C. M. Kremer, A. Eremiev, B. Negash, D. D. Wiggan, M. A. Nakatsuka, K. L. Sangwon, S. Neifert, H. A. Khan, A. V. Save, A. Palla, E. A. Grin, M. Hedman, M. Nasir-Moin, C. X. Liu, L. Y. Jiang, M. A. Mankowski, D. L. Segev, Y. Aphinyanaphongs, H. A. Riina, J. G. Golfinos, D. A. Orringer, D. Kondziolka, E. K. Oermann, *Medical Large Language Models are Vulnerable to Data-poisoning Attacks*, Nature Medicine, 2025. [journal] [code]
16. R. Guennoun[†], **A. Alyakin**[†], H. Higushi, S. Demehri, *Commensal HPVs Have Evolved to Be More Immunogenic Compared with High-Risk α -HPVs*, Vaccines, 2024. [journal]
17. J. Lee, M. A. Ruiz-Cardozo, R. P. Patel, S. Javeed, R. S. Lavandi, C. Newsom-Stewart, **A. Alyakin**, C. A. Molina, N. Agarwal, W. Z. Ray, M. Santacatterina, B. H. Pennicooke, *Clinical Prediction for Surgical versus Nonsurgical Interventions in Patients with Vertebral Osteomyelitis and Discitis*, Journal of Spine Surgery, 2024. [journal]
18. **A. A. Alyakin**, J. Agterberg, H. S. Helm, and C. E. Priebe, *Correcting a Nonparametric Two-sample Graph Hypothesis Test for Graphs with Different Numbers of Vertices with Applications to Connectomics*, Applied Network Science, 2024. [journal] [arXiv] [code]
19. J. Chung[†], B. Varjavand[†], J. Arroyo, **A. Alyakin**, J. Agterberg, M. Tang, J. T. Vogelstein, C. E. Priebe, *Valid Two-Sample Graph Testing via Optimal Transport Procrustes and Multiscale Graph Correlation with Applications in Connectomics*, Stat, 2022. [journal] [arXiv] [code]
20. M. Powell, C. Clark, **A. Alyakin**, J. T. Vogelstein, B. Hart, *Exploration of Residual Confounding in Analyses of Associations of Metformin Use and Outcomes in Adults With Type 2 Diabetes*, JAMA Network Open, 2022. [journal] [arXiv]
21. F. Rahman, N. Finkelstein, **A. Alyakin**, N. A. Gilotra, J. Trost, S. P. Schulman, S. Saria, *Using Machine Learning for Early Prediction of Cardiogenic Shock in Patients with Acute Heart Failure*, Journal of the Society for Cardiovascular Angiography & Interventions, 2022. [journal] [arXiv]
22. K. Marchisio, Y. Park, A. Saad-Eldin, **A. Alyakin**, K. Duh, C. Priebe, P. Koehn, *An Analysis of Euclidean vs. Graph-Based Framing for Bilingual Lexicon Induction from Word Embedding Spaces*, Findings of the Association for Computational Linguistics: EMNLP 2021. [journal] [arXiv] [code]

PREPRINTS

1. J. E. Markert, N. K. Goff, **A. Alyakin**, J. V. Lee, K. Vishwanath, E. Yang, A. Kapoor, J. J. Choi, R. K. Steele, A. Schulze, S. F. Chen, J. Lee, E. K. Oermann, *Computer vision in clinical medicine: an LLM-powered large-scale systematic review*, submitted. [preprint]
2. M. Nakatsuka, M. Murata, D. D. Wiggan, Z. E. Morakinyo, C. Cubells, **A. Alyakin**, E. K. Oermann, *GPT-4-Assisted versus Human Detection of Intraoperative Adverse Events from Spine Surgery Operative Reports*.
3. A. Arias, J. Markert, **A. Alyakin**, A. Kapoor, D. Orringer, E. K. Oermann, *Predictors of Neurosurgical Academic Placement and Research Productivity: A Systematic Review*, submitted.
4. Y. Hong, S. Son, H. Jung, K. Vishwanath, **A. Alyakin**, J. V. Lee, E. K. Oermann, M. Sung, *Learning to Ignore: Enhancing Medical QA robustness under Unseen Distractors*, submitted.
5. A. K. Kapoor, **A. Alyakin**, J. V. Lee, E. Yang, K. Vishwanath, A. M. Schulze, J. Lee, Y. Aphinyanaphongs, H. Riina, J. A. Frontera, E. K. Oermann, *Large Language Models for Predicting Functional Outcomes after Acute Ischemic Stroke*, submitted. [arXiv]

6. **A. Alyakin**, "All You Need" Is Not All You Need for a Paper Title: On the Origins of a Scientific Meme, 2025. [arXiv] [code]
7. L. Y. Jiang[†], A. Chen[†], X. Han, X. C. Liu, R. Dua, K. Eaton, F. Wolff, R. Steele, J. Zhang, **A. Alyakin**, Q. Pan, Y. Chen, K. L. Sangwon, D. A. Alber, J. Stryker, J. V. Lee, Y. Aphinyanaphongs, K. Cho, E. K. Oermann, *Generalist Foundation Models Are Not Clinical Enough for Hospital Operations*, 2025. [arXiv]
8. K. Vishwanath, **A. Alyakin**, J. V. Lee, A. Anand, D. A. Alber, E. K. Oermann, *Medical Large Language Models are Easily Distracted*, 2025. [arXiv] [code] [dataset]
9. C. Hang, R. Deng, L. Y. Jiang, Z. Yang, D. A. Alber, **A. Alyakin**, E. K. Oermann, *BPQA Dataset: Evaluating How Well Language Models Leverage Blood Pressures to Answer Biomedical Questions*, 2025. [arXiv]
10. **A. Alyakin**, Y. Qin, and C. E. Priebe, *LqRT: Robust Hypothesis Testing of Location Parameters using Lq-Likelihood-Ratio-Type Test in Python*, 2019. [arXiv] [code]

PRESENTATIONS

1. **A. Alyakin**, J. Stryker, D. A. Alber, J. V. Lee, E. Leuthardt, D. Kondziolka, E. K. Oermann, *CNS-Obsidian: A Neurosurgery-Tailored Vision-Language Model Fine-Tuned Exclusively on Scientific Publications*, Oral Presentation at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [journal] [poster]
2. X. Han[†], **A. Alyakin**[†], S. Ciprut, C. Lapierre, J. Stryker, A. Lowe, J. Golfinos, D. Kondziolka, E. K. Oermann, *Neuro Data Hub: A New Approach For Streamlining Medical Clinical Research*, Oral Presentation at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [poster]
3. K. Vishwanath, **A. Alyakin**, M. Ghosh, J. V. Lee, D. A. Alber, K. Sangwon, D. Kondziolka, E. K. Oermann, *Evaluating the Performance and Fragility of Large Language Models on Neurosurgical Board-like Questions*, Mini Oral Presentation at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [journal] [poster]
4. **A. Alyakin**, *Computational Optimization Of Laser Interstitial Thermal Therapy Parameters For Brain Tumor Ablation: A Thermodynamics-Based Simulation Approach*, Mini Oral Presentation at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [journal] [poster]
5. **A. Alyakin**, J. Stryker, S. Singh, D. A. Alber, A. Save, D. Kurland, C. Orillac, A. A. Valliani, S. Neifert, D. Lau, I. Laufer, P. A. Rozman, E. T. Hidalgo, H. Riina, L. Snyder, E. C. Leuthardt, D. Kondziolka, E. Oermann, *Automated Generation And Human Evaluation Of Neurosurgical Board Examination And Self-Assessment Questions*, Mini Oral Presentation at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [journal] [poster]
6. A. Y. Fu, A. A. Valliani, **A. Alyakin**, D. Kondziolka, E. K. Oermann, *Tabular Foundation Models Predict Clinical Outcomes in Skull Base and Cerebrovascular Surgery*, Mini Oral Presentation at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [journal] [poster]
7. A. Save, **A. Alyakin**, J. Stryker, D. A. Alber, J. Zhang, G. Rosenbaum, S. Singh, E. Leuthardt, D. Kondziolka, E. K. Oermann, *A Platform for Serving Large Language Models for Neurosurgical Diagnostic Support*, 93rd American Association of Neurological Surgeons Annual Scientific Meeting, 2025.
8. K. L. Sangwon, **A. Alyakin**, D. Kurland, E. Leuthardt, D. Kondziolka, E. K. Oermann, *A Generalizable Pipeline for Building an Extensive Domain-Specific Dataset from a Medical Journal - Neurosurgery Edition*, Oral Presentation at the 74th Annual Meeting of the Congress of the Neurological Surgeons, 2024.

9. **A. Alyakin**, D. Kurland, D. Alber, K. Sangwon, D. Li, A. Tsirigos, E. Leuthardt, D. Kondziolka, E. Oermann, *CNS-CLIP: Transforming a Neurosurgical Journal into a Multimodal Medical Model*, Oral Presentation at the 74th Annual Meeting of the Congress of the Neurological Surgeons, 2024.

POSTERS

1. A. K. Kapoor, **A. Alyakin**, E. Yang, A. S. Korpus, A. Schulze, J. E. Markert, J. A. Frontera, E. K. Oermann, *Predicting Functional Outcomes After Stroke Using Large Language Models*, Digital Poster at the 94th American Association of Neurological Surgeons Annual Scientific Meeting, 2026.
2. J. Choi, **A. Alyakin**, A. Schulze, S. F. Chen, E. K. Oermann, *Large Language Model-assisted Systematic Review of Generative AI in Neurosurgery*, Digital Poster at the 94th American Association of Neurological Surgeons Annual Scientific Meeting, 2026.
3. **A. Alyakin**, J. Stryker, J. V. Lee, R. Feng, T. Hollon, D. Kondziolka, E. K. Oermann, *AI-Powered Pipeline Transforms Neurosurgical Articles into High-Quality Graphical Abstracts*, Digital Poster at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [poster]
4. **A. Alyakin**, J. V. Lee, H. Riina, E. K. Oermann, *Computational Optimization Of Endovascular Coil Deployment Parameters For Aneurysm Treatment: A Physics-Based Simulation Approach*, Digital Poster at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025.
5. S. Singh, J. Zhang, **A. Alyakin**, E. K. Oermann, *GRPO Versus SFT: Optimizing Large Language Models For Neurosurgical Decision Support*, Physical Poster at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [poster]
6. A. Kapoor, **A. Alyakin**, L. Blinder, A. Korpus, E. Yang, E. Yang, R. J. Steele, A. M. Schulze, J. A. Frontera, E. K. Oermann, *Large Language Models Predict Modified Rankin Scale After Acute Ischemic Stroke*, Digital Poster at the 75th Annual Meeting of the Congress of the Neurological Surgeons, 2025. [poster]
7. G. R. Rosenbaum, L. Y. Jiang, I. Sheth, J. Stryker, **A. Alyakin**, D. A. Alber, N. K. Goff, Y. J. F. Kwon, J. E. Markert, M. Nasir-Moin, J. M. Niehues, K. L. Sangwon, E. Yang, E. K. Oermann, *MedG-KRP: Medical Graph Knowledge Representation Probing*, Findings of the 4th Machine Learning for Health symposium (ML4H), 2024. [arXiv]
8. K. L. Sangwon, D. Kurland, **A. Alyakin**, D. Kondziolka, E. K. Oermann, *Seven Decades Of Change: Tracing The Evolution Of Neurosurgery Through Lexical Analysis Of Neurosurgery Publications Of The CNS (1955-2024)*, Digital Poster at the 74th Annual CNS Meeting, 2024.

NON-REFEREED

1. **A. Alyakin**, A. Liu, C. Sen, E. K. Oermann, *Suboccipital Craniectomy with C1 Laminectomy for the Resection of Cervicomedullary Hemangioblastoma*, CNS-Nexus Case, 2026.
2. J. Choi, **A. Alyakin**, K. Sangwon, E. K. Oermann, *Repeat Endoscopic Fenestration of a Recurrent Symptomatic Left Arachnoid Cyst with Midline Shift*, CNS-Nexus Case, 2026. [case]

TALKS

Congress of Neurological Surgeons

CNS 2025 Special Symposium 3: Data Science and AI for Neurosurgeons

Oct 11th, 2025

Vision-Language Models

2025 Artificial Intelligence in Neurosurgery: A Practical Course

May 31st, 2025

How Can We Build Foundation Models with Neurosurgical Data

Global AI Frontier Lab

Global AI Frontier Lab: Seminar Series

Jun 16th, 2025

Data Quality is Multidimensional: Product-Market-Data Fit in AI and Neurosurgery

AWARDS

Johns Hopkins University

Applied Mathematics & Statistics Prize for Outstanding Master's Research

2020

Applied Mathematics & Statistics Achievement Award

2019

Undergraduate Departmental Honors with Thesis, Computer Science

2019

Undergraduate Departmental Honors, Applied Mathematics & Statistics

2019

Undergraduate General Honors

2019

Whiting School of Engineering Dean's List (8/8 Semesters)

2015-2019

TASIS The American School In Switzerland

2015 Shah Akbar Khan Award for Excellence in Mathematics

2015

2015 AP Scholar with Distinction

2015

Valedictorian

2015

REVIEW

Neurosurgery

Reviewer

[2024, present]

4 articles

Quantitative Imaging in Medicine and Surgery

Reviewer

[2024, present]

2 articles

Scientific Reports

Reviewer

[2025, present]

3 articles

npj: Digital Medicine

Reviewer

[2025, present]

3 articles

BMC Neurology

Reviewer

[2025, present]

1 article

IEEE Transactions on Signal and Information Processing over Networks

Reviewer

[2026, present]

1 article

Frontiers in Artificial Intelligence

Reviewer

[2026, present]

1 article

PLoS ONE

Reviewer

[2026, present]

2 articles

MENTORING

Joanne Choi

Visiting Medical Student Researcher at NYU OLAB

[2025, 2025]

1 poster (AANS 2026)

1 *CNS-Nexus* case

Anjali Kapoor

Post-baccalaureate Researcher at NYU OLAB

[2025, Present]

1 manuscript (Journal of Neuro-Oncology)

2 manuscript submitted

2 posters (CNS 2025, AANS 2026)

Krithik Vishwanath

Visiting Undergraduate Researcher at NYU OLAB

[2024, Present]

3 manuscript published

Nature Medicine

Neurosurgery

BMJ Digital Health & AI

Churchill scholar

Shrutika Singh

Undergraduate Researcher at NYU OLAB

[2024, 2025]

1 manuscript published (Scientific Reports)

1 oral presentation (CNS 2025)

1 poster (CNS 2025)

Gabriel Rosenbaum

High School Student Researcher at NYU OLAB

[2024, 2024]

1 poster (ML4H 2024)

TEACHING

Congress of Neurological Surgeons

Faculty

CNS 2025 Special Symposium 3: Data Science and AI for Neurosurgeons

Oct 2025

2025 Artificial Intelligence in Neurosurgery: A Practical Course

May 2025

Teaching Assistant

CNS 2024 Special Symposium 3: Data Science and AI for Neurosurgeons

Sep 2024

Johns Hopkins University

Teaching Assistant

580.475 Biomedical Data Science

Fall 2019

553.430/630 Introduction to Statistics

Spring 2019

553.436/636 Data Mining

Fall 2018

SOCIETIES

Congress of Neurological Surgeons

Medical Student Member

[2024, present]

CNS Quarterly Member Spotlight

Summer 2025

Sergeant-At-Arms

CNS 2025

American Association of Neurological Surgeons

Medical Student Member

[2021, present]